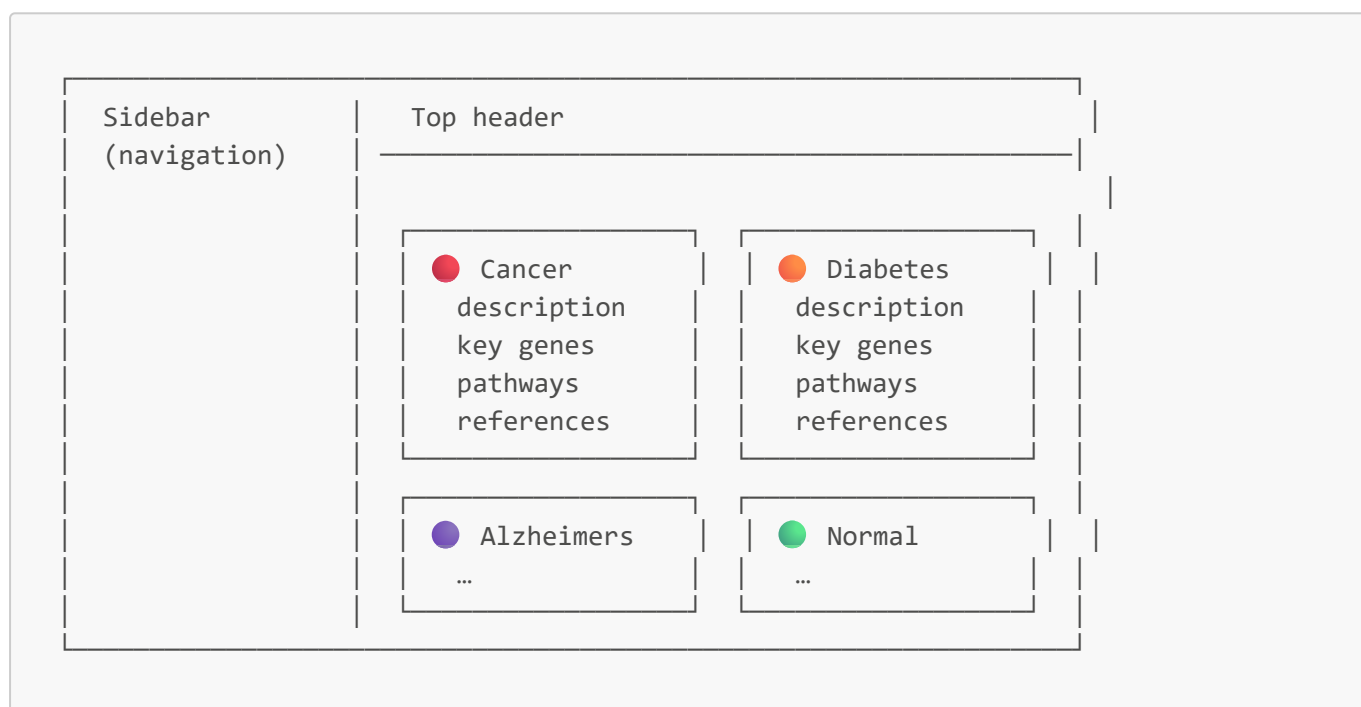


Knowledge Base Page — Complete Component Guide

A deep-dive into the **Knowledge Base** page of GenomelQ. For each component this document covers:

-  **What it is** (plain-language description)
-  **What it tells you** (how to read the output)
-  **How it works under the hood** (the algorithm)
-  **How to use it** (the user flow)
-  **Practical tips** (when it's useful, when to be cautious)

Page Layout Overview



The Knowledge Base page is a clean **2-column responsive grid** of disease cards. On mobile each card stacks full-width.

1. Disease Cards (Four Total)

What it is

A vertical card per disease class providing reference information about that disease — its biology, genetic markers, pathways, and external links.

What it tells you

Each card has six sections from top to bottom:

Section	Meaning
Top color band	A 4-px solid border color-coded to the disease (red Cancer, amber Diabetes, purple Alzheimers, emerald Normal).
Icon + name + short description	Header row: emoji, class name, one-line summary.
Long description paragraph	Two-to-three-sentence biological summary of the disease class.
Key genes (chips)	Marker genes for the class — BRCA1 , BRCA2 , TP53 for Cancer; INS , INSR , GCK for Diabetes; etc.
Pathways (chips)	Biological pathways involved — <i>DNA damage response</i> , <i>Insulin signalling</i> , <i>Amyloid-β processing</i> , etc.
References (links)	NCBI Gene URLs you can click through to verify and dig deeper.

⚙️ How it works under the hood

The cards are populated from the API endpoint `GET /diseases`, which is served by `src/api/service.py → diseases_payload()`. That function returns `list[dict]` from the in-memory dictionary `DISEASE_INFO` defined in `src/core/knowledge.py`:

```
DISEASE_INFO: dict[str, DiseaseInfo] = {
    "Cancer": DiseaseInfo(
        name="Cancer",
        icon="🔴",
        color="#ef4444",
        short_description="Markers of oncogenic gene expression detected.",
        long_description="Cancer is characterised by uncontrolled cell growth
    ...",
        key_genes=("BRCA1", "BRCA2", "TP53", "EGFR", "MYC", "KRAS", "PTEN"),
        pathways=("DNA damage response", "Cell cycle", ...),
        references=("https://www.ncbi.nlm.nih.gov/gene/672", ...),
    ),
    ...
}
```

The Alpine front end fetches this once at page load (in `genomeApp().fetchKnowledge()`) and stores it in `state.knowledge`. The Knowledge Base page iterates over the array with `<template x-for>`.

📖 How to use it

1. Click  **Knowledge Base** in the sidebar.
2. Browse the four cards.
3. Click any reference link to open the corresponding NCBI Gene page in a new tab.

💡 Practical tips

- **The data is curated**, not auto-generated — this is a deliberate design choice. The same **DISEASE_INFO** is reused by:
 - The **PDF report** (renders the predicted disease's metadata).
 - The **Analyze verdict card** (uses the colors and icons).
 - The **RAG Copilot knowledge base** (the disease and gene facts are augmented and indexed for retrieval).
 - This page is **read-only**. To add a new disease class, edit `src/core/knowledge.py` and restart the server. (Heads up: the actual classifier still only knows the four classes it was trained on — adding a card doesn't add a class.)
-

2. 🧬 Key Gene Chips

🎯 What it is

Small monospace pill labels under "Key genes" — one per gene that's a marker for the disease class.

💬 What it tells you

Each chip is a **gene symbol** (HGNC convention, uppercase). Chips do not currently link to external pages.

⚙️ How it works under the hood

Pulled from the `key_genes` tuple in **DISEASE_INFO**. They're read-only metadata — the classifier doesn't directly use this list, but it's the same list the **Copilot** references when answering gene-related questions.

💡 Practical tips

- For **deeper exploration**, copy the gene symbol and:
 - Click the matching reference URL (NCBI Gene).
 - Or paste it into the **Copilot** with a question like "What does *BRCA1* do?"
 - The genes are listed in roughly **descending order of disease relevance** (e.g., BRCA1, BRCA2 first for Cancer).
-

3. 🗺️ Pathway Chips

🎯 What it is

Small label pills under "Pathways" — one per **biological pathway** the disease's marker genes are involved in.

💬 What it tells you

Each chip is a pathway or biological process — for example, *DNA damage response*, *Cell cycle*, *PI3K/AKT*, *Insulin signalling*, *Amyloid-β processing*.

⚙️ How it works under the hood

Stored in the `pathways` tuple of each **DiseaseInfo**. Like the genes, they're read-only metadata.

💡 Practical tips

- These are **simplified pathway labels** for educational context — they aren't the full Reactome / KEGG identifiers.
- Use them as **search terms** when looking up disease biology in textbooks or PubMed.

4. Reference Links

What it is

Clickable URLs at the bottom of each card pointing to external genomic resources.

What it tells you

Each link is an **NCBI Gene** entry for a key marker. NCBI Gene is the canonical reference for gene metadata (chromosomal location, expression, links to literature).

How it works under the hood

Plain HTML `<a>` tags rendered from the `references` tuple, with `target="_blank"` so they open in a new tab.

How to use it

Click a link → opens **`ncbi.nlm.nih.gov/gene/...`** in a new browser tab. From there you can:

- Read the official gene description
- See chromosomal location
- Access OMIM clinical synopses
- Download FASTA or RefSeq sequences

Practical tips

- The links are **stable** — NCBI gene IDs don't change.
- To add **new references** (e.g., OMIM, Ensembl), just add URLs to the `references` tuple in `src/core/knowledge.py`.

The Knowledge Base Flow (End-to-End)

```
Page load (init() in app.js)
  ↓
[GET /diseases]
  ↓
Backend: src/api/service.py → diseases_payload()
  |
  └─ core/knowledge.py → list_diseases()
      ↓
      list of DiseaseInfo objects → JSON
        ↓
Alpine state.knowledge = [...]
  ↓
User clicks " Knowledge Base" sidebar item
```

```

↓
<template x-for="d in knowledge">
  ↓ render card with:
  ↓ - top border in d.color
  ↓ - icon + name + short_description
  ↓ - long_description paragraph
  ↓ - key_genes chips
  ↓ - pathways chips
  ↓ - references as clickable links

```

Total time: **<10 ms** to fetch the JSON; rendering is essentially instant.

Cheat Sheet — How the Knowledge Base Connects to the Rest of GenomelQ

```

core/knowledge.py
├─ seeds disease icons + colors → used in Analyze verdict card
├─ seeds disease description → used in PDF report
├─ seeds key_genes + pathways → used in Knowledge Base page
├─ augmented in rag/knowledge_base.py
│   └─ indexed by FAISS for the Copilot retrieval

```

The same data drives **four different UI surfaces** plus the LLM's grounding context. Adding one entry once propagates everywhere.

Extending the Knowledge Base

If you want to add a new gene fact:

1. Open `src/rag/knowledge_base.py`.
2. Add a new `Fact(...)` to `GENE_FACTS`.
3. Delete `models/rag_index.faiss` (so it rebuilds with the new fact).
4. Restart the server.

If you want to add a new disease class to the **UI cards** (note: classifier won't recognize it):

1. Open `src/core/knowledge.py`.
2. Add a new entry to `DISEASE_INFO`.
3. Refresh the page — the new card appears.

For details on retraining the actual classifier on a new class, see [DOCS.md](#) section 10.

Related Documents

- [README.md](#) — high-level project overview
- [DOCS.md](#) — comprehensive technical documentation

- [Guide/COPILLOT_PAGE_GUIDE.md](#) — uses the same knowledge base for retrieval
 - [Guide/ANALYZE_PAGE_GUIDE.pdf](#) — also pulls disease metadata from [core/knowledge.py](#)
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Part of the GenomelQ per-page documentation series.